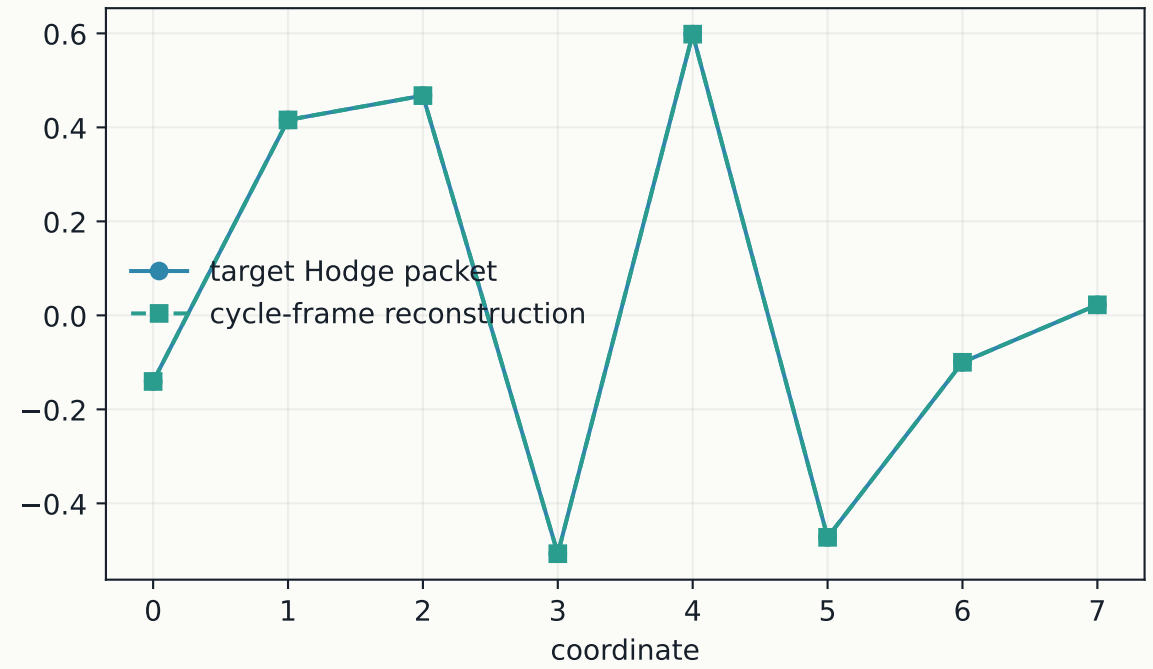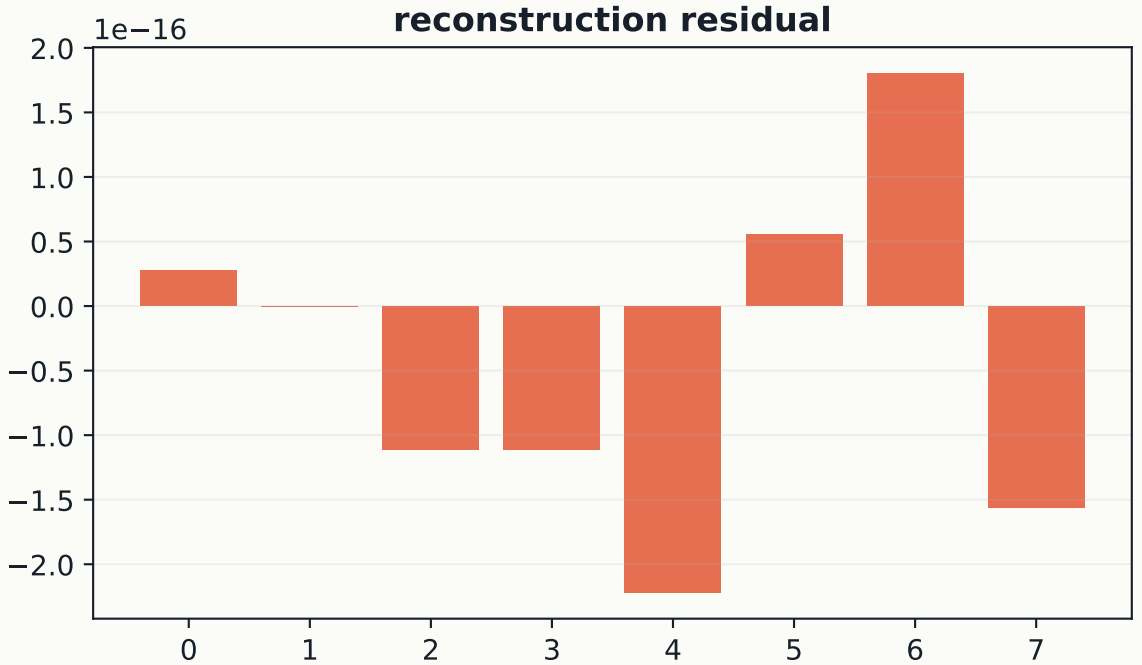Exact synthesis is ordinary finite-dimensional frame reconstruction